

■ BUILD MANIFEST

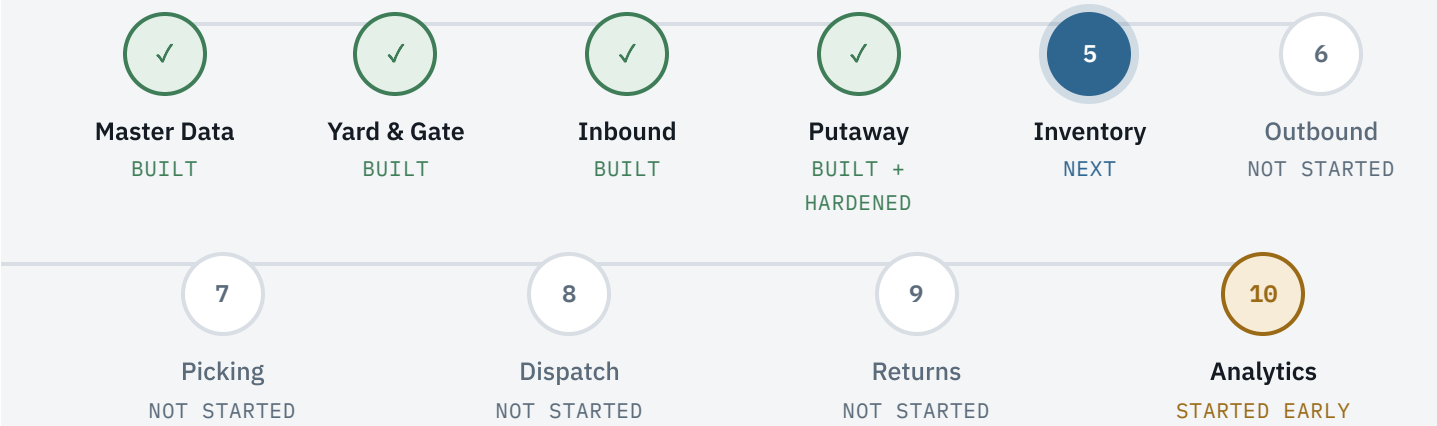
WMS Roadmap

Cover the basics of every module first, then come back and deepen each one — rather than gold-plating one module before the rest exist at all. This stretch also ran a deliberate "hardening phase" — going back over Putaway's live behavior to find and fix real bugs before adding more scope.

Repo VasuRao97/WMS Updated 2026-09-06 Latest commit 6b3e88e9

Module build order

4 of 10 shipped · Analytics started early, out of sequence



What's landed since the last update (Aug 27 → Sep 6)

11 sessions' worth

- Putaway — the module itself, live-tested and hardened

Core bin-suggestion logic, scan-driven claim/complete execution, Multi-SKU Lane Exception workflow, then a live-testing pass caught and fixed three real bin-suggestion bugs (pending-reservation blind spot, lane-fullness preference, mirrored-flank merging).

BUILT
- Pallet consolidation, Drive-in split, operator-assignment fairness

BUILT

"Marrying" loose cases onto a pallet before Putaway; Drive-in got its own physically-correct bin-suggestion strategy (whole-column, deepest-tier-first); a live "who goes next" fairness recommendation with Supervisor→Manager escalation.

Analytics module begins

BUILT

Operator productivity at the Pallet level (marrying time + putaway time, split per operator) plus abandoned-claim flagging — no new schema, fully derived from existing data.

Insights — Storage Utilization by ABC Class

BUILT

A standalone reporting page: of the rack lanes actually holding stock per ABC class, how much of that lane's depth is occupied.

Locations/Bins: real 3D Plan View

BUILT

A true WebGL camera (Three.js) alongside the existing 2D top-down view — whole-warehouse footprint blocks with a slicer to drill into full per-bin detail, plus occupancy-by-Category/Class overlays, click-to-inspect (now with SKU details), a Level toggle, and smooth camera auto-focus.

Putaway Simulation sandbox

BUILT

A dedicated sandbox warehouse replays the real, unmodified suggestBin() algorithm against synthetic SKUs as a speed-adjustable step-by-step animation, through the same Plan View.

Putaway location override + discrepancy highlighting

BUILT

A per-company toggle lets an operator complete at a different real bin instead of a hard block; any mismatch is flagged on the task queue and in a dedicated Supervisor+ review list.

ABC velocity reassessment (Topic 1)

BUILT

A real monthly job re-derives each SKU's A/B/C/D class PER WAREHOUSE from actual trailing dispatch quantity — replacing blind trust in a manually-typed/imported class.

Dock-relative Putaway placement (Topic 2)

BUILT

A new per-warehouse Dock Configuration now drives fast movers toward the Outbound end and slow movers away from it — live-verified to actually flip direction between a U-shape and I-shape layout.

Immediate candidates — next session

pick one

01

02

Reslotting/consolidation engine

The daily suggestion engine flagged since Topic 1 — which SKUs are misplaced, which bins/pallets can be consolidated. Schema's ready and now genuinely unblocked (Topic 2's placement rules exist), but needs its own design pass first.

needs design pass

Inventory (on-hand view)

Next in the stated build order. There is still no screen anywhere showing "what's on hand at Location X," even though real ledger data exists now. Comparatively light — mostly a read/report over data already being written.

03

Outbound order maker

The next real module after Inventory — destination + vehicle capacity checks, triggering a pick list. Benefits from Inventory existing first.

04

Dock-out → Gate-out signal

Flagged twice now — a notification once documentation/paperwork time after dock-out is up, before Gate Out is actually allowed. Needs its own design conversation, same care as Dock Scheduling.

needs design pass

05

Dock + Staging + Yard visualizer

Paused mid-conversation with four open questions still unanswered (positional data on Dock Doors/Yard Slots, live vs. static, where it lives, whether it links to the real Locations Plan View). Needs a spatial-data pass before anything else.

needs design pass

Deferred — lower priority

don't build unprompted

Self-service driver check-in

PARKED

Schema-only (SelfCheckInRequest) — still needs its own align-before-coding pass; drivers have no login/User account anywhere in this system.

Real GS1 / unique-barcode parsing ("Reading B")

PARKED

The tyres/FMCG-case problem. Supervisor-approve fallback makes those categories usable without it today.

Camera-based scanning

PARKED

Hardware-scanner (keyboard-wedge) input works everywhere today; no camera/decode library wired up.

Real SMS / Email / WhatsApp / Voice providers

PARKED

Every channel is still a stub that only logs. MSG91 and Exotel were the research leads — neither chosen or wired up.

Dock Scheduling

PARKED

Zero design/schema. Only its future output (a dock number, a driver's assigned dock) has anywhere to land today.

Ground/Stillage's own Putaway logic, and ASRS's own strategy

PARKED

Both still inherit rack-generic (SPR) behavior — "let's finish racked first," ASRS explicitly parked for later.

No cancel/undo paths yet

PARKED

A mistaken Gate In, an un-completable Putaway task, and correcting an already-completed mis-putaway all have no correction flow today.

Batch/Lot + expiry capture, and closing a short receipt

PARKED

Both resurfaced twice in competitor research (feeds a future FEFO policy) — never picked up.

MHE travel-time-based placement

PARKED

DockLocationDistance (real meters) and Equipment loaded/unloaded speed are schema-only — no data-entry tooling, no consuming logic. Today's placement uses a flat Level rule instead.

Blacklist enforcement at Gate In, Yard analytics dashboard, WMS/TMS integration

PARKED

Explicitly parked — don't propose as quick wins. (Distinct from the real Analytics module now underway.)

Platform & ops: SUPER_ADMIN creation, permission matrix, invite-link flow, Inventory Control Policy, Opening Balance, dispatch-proximity calc, real SAP/ERP integration, cloud deployment

PARKED

All fully deferred — no schema or design work done.

How to use this file: at the end of a session, update the module status and the immediate-candidates list before committing — that's what keeps this useful for picking up work cold, instead of re-deriving it from CLAUDE.md's full history each time.